

Business Requirements Document (BRD)

Cloud-Based Chilling Centre Milk Collection & Device Integration System

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Status: Draft

1. Purpose

The purpose of this system is to digitize and automate the **milk reception and collection process at a Chilling Centre**.

The system will capture milk quantity and quality electronically by integrating with:

- Electronic weighing scales
- Milk analysers

The application will be hosted in the cloud and accessed through a web browser, while the physical devices at the chilling centre will be accessed through a **Local Device Gateway**.

The system shall also implement **Role-Based Access Control (RBAC)** to ensure that users can access only the functions and data appropriate to their responsibilities.

The target process is:

Milk Arrival → Source/Vehicle Identification → Quantity Measurement → Quality Measurement → Automatic Data Capture → Validation → Accept/Reject → Receipt → Cloud Storage → Dashboard & Reports

2. Business Objectives

The system shall:

1. Eliminate manual recording of milk quantity and quality.
2. Automatically capture weight from electronic weighing scales.
3. Automatically capture quality readings from milk analysers.
4. Support RS232 and Bluetooth device connectivity.
5. Provide a cloud-hosted web application.
6. Allow operators to work through a browser.
7. Provide local device connectivity through a Local Device Gateway.

8. Reduce data-entry errors.
 9. Provide real-time visibility of milk reception.
 10. Maintain a complete audit trail.
 11. Support temporary internet outages.
 12. Support multiple chilling centres.
 13. Provide configurable RBAC.
 14. Restrict sensitive operations to authorized users.
 15. Provide a scalable architecture supporting different device manufacturers and models.
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3. Scope

3.1 In Scope

Chilling Centre Operations

- Milk reception
- Source identification
- Vehicle/tanker identification
- Milk quantity capture
- Milk quality capture
- Quality validation
- Milk acceptance
- Milk rejection
- Milk hold
- Receipt generation
- Daily dashboard
- Reports

Device Integration

- Weighing scale integration
- Milk analyser integration
- RS232 connectivity
- Bluetooth connectivity
- Device configuration
- Device status monitoring
- Device communication logging
- Automatic reconnection
- Device data parsing
- Manual fallback

Application

- Cloud-hosted web application
- Authentication
- Role-Based Access Control

- User management
- Centre-level access control
- Module-level access control
- Action-level authorization
- Master data
- Audit trail
- Reports
- Dashboard
- Offline transaction handling

4. Out of Scope

The initial version will not include:

- Farmer payment management
- Full dairy ERP
- Dairy production management
- Sales management
- Accounting
- GPS tracking
- Route optimization
- Advanced laboratory management
- AI/GenAI
- Predictive analytics

These may be added in future phases.

5. Users and Roles

The system shall support configurable roles.

The following roles are recommended for the initial implementation:

Role	Primary Responsibilities
System Administrator	Complete system configuration and administration
Centre Administrator	Manage configuration for assigned chilling centre
Chilling Centre Manager	Monitor operations, reports and approvals
Milk Reception Operator	Record and process milk reception
Quality Operator	Perform/verify quality operations
Device/Technical Support	Configure and troubleshoot devices
Management / Viewer	View dashboards and reports

Roles shall be configurable rather than hard-coded wherever practical.

6. Role-Based Access Control (RBAC)

RBAC shall be a **mandatory core security requirement**.

Access shall be determined by:

User → Role → Permissions → Resource

The system shall not grant access based solely on the user's login.

7. RBAC Permission Model

Permissions shall be defined at multiple levels.

7.1 Module Level

Examples:

- Dashboard
- Milk Reception
- Sources
- Vehicles
- Quality
- Devices
- Reports
- Users
- Roles
- Configuration
- Audit Logs

7.2 Action Level

Each module may have actions such as:

- View
- Create
- Edit
- Delete
- Approve
- Reject
- Override

- Export
- Configure
- Test Device

For example:

Milk Reception

- View
- Create
- Edit
- Cancel
- Approve
- Reject
- Hold
- Override

8. Example RBAC Matrix

Function	System Admin	Centre Admin	Manager	Operator	Quality	Device Support	Viewer
Dashboard	✓	✓	✓	✓	✓	✓	✓
Milk Reception	✓	✓	✓	✓	View	View	View
Create Transaction	✓	✓	✓	✓	-	-	-
Edit Transaction	✓	✓	✓	Limited	-	-	-
Accept Milk	✓	✓	✓	✓*	✓	-	-
Reject Milk	✓	✓	✓	Limited	✓	-	-
Override Decision	✓	✓	✓	-	Limited	-	-
Sources	✓	✓	View	View	View	-	View
Vehicles	✓	✓	View	View	View	-	View
Quality Rules	✓	✓	View	-	View	-	-
Device Configuration	✓	✓	-	-	-	✓	-
Device Testing	✓	✓	-	-	-	✓	-
Reports	✓	✓	✓	Limited	✓	Limited	✓
User Management	✓	✓*	-	-	-	-	-
Role Management	✓	-	-	-	-	-	-

Function	System Admin	Centre Admin	Manager	Operator	Quality	Device Support	Viewer
Audit Logs	✓	✓	✓		- Limited	✓	-
System Configuration	✓	✓*	-	-	-	-	-

* Access shall be configurable according to the organization's approval policy.

9. Centre-Level Access Control

The system shall support multiple chilling centres.

A user may be assigned to:

- One chilling centre
- Multiple chilling centres
- All chilling centres

Example:

User: Operator001

Role:
Milk Reception Operator

Assigned Centre:
Chilling Centre - Bangalore

Access:

Milk Reception	✓
Reports	✓
Other Centres	X
User Management	X
Device Configuration	X

A manager may have access to multiple centres.

Management users may have organization-wide visibility.

10. Authentication

The system shall provide secure user authentication.

Minimum requirements:

- Username/email
- Password
- Secure password policy
- Account activation/deactivation
- Session management
- Logout
- Password reset

The architecture should allow future support for:

- OTP
- Microsoft/Google SSO
- MFA
- Enterprise identity providers

11. Authorization

Authentication determines:

Who are you?

Authorization determines:

What are you allowed to do?

Every protected API and application operation shall perform authorization checks.

RBAC shall be enforced at both:

- Frontend/UI level
- Backend/API level

Frontend restrictions alone shall not be considered sufficient security.

12. Privileged Operations

The following operations shall require explicit permissions:

- Changing quality limits
- Changing device configuration
- Adding/removing devices
- Manual quantity entry
- Manual quality entry
- Accepting rejected milk

- Rejecting otherwise acceptable milk
- Transaction cancellation
- Transaction modification
- User creation
- Role creation
- Permission assignment
- Exporting sensitive reports

13. Approval / Override Controls

For sensitive operations, the system may require approval from a higher-level role.

Example:

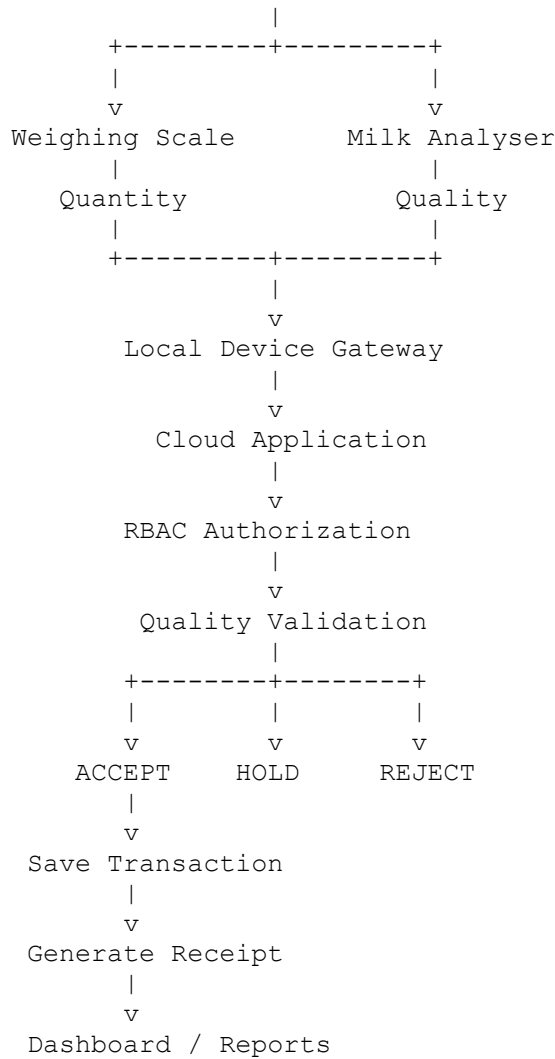
```
Operator
  |
  | Milk Quality Outside Limit
  v
System → HOLD
  |
  v
Manager Review
  |
  +----- ACCEPT
  |
  +----- REJECT
```

The override shall record:

- Original status
- New status
- User
- Date/time
- Reason
- Approval information

14. High-Level Business Process

```
MILK VEHICLE ARRIVES
  |
  v
User Authentication
  |
  v
Source / Vehicle Selection
  |
  v
Milk Reception
```



15. Cloud Web Application

The core application shall be a cloud-hosted web application.

Users shall access it through a standard browser.

The application shall provide:

- Login
- Dashboard
- Milk Reception
- Source Management
- Vehicle Management
- Device Management
- Quality Configuration
- Reports
- User Management
- Role Management

- Permission Management
- Audit Logs

The main application should not require local installation on every operator computer.

16. Local Device Gateway

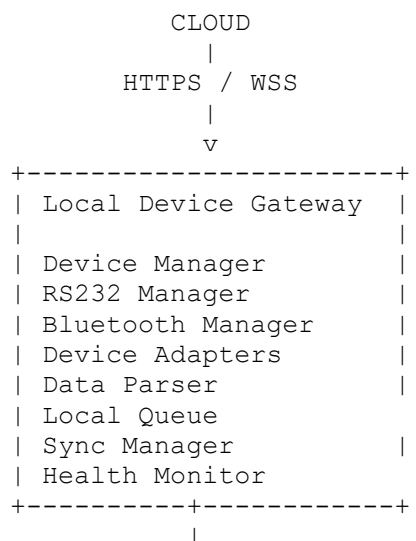
A lightweight **Local Device Gateway** shall be installed at each chilling centre where physical devices are located.

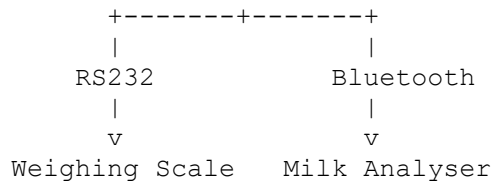
Its purpose is to communicate with local equipment and securely exchange data with the cloud application.

The gateway shall:

- Detect configured devices
- Connect to devices
- Read device data
- Parse device messages
- Normalize readings
- Send readings to the cloud
- Monitor device connectivity
- Reconnect automatically
- Maintain local logs
- Buffer readings during temporary internet outages
- Synchronize data with the cloud

17. Local Device Gateway Architecture





18. Weighing Scale Integration

The system shall support electronic weighing scales through:

- RS232
- Bluetooth

Both shall be mandatory supported communication options.

19. Weighing Scale — RS232

The Local Device Gateway shall support RS232/Serial communication.

Configurable parameters:

- COM Port
- Baud Rate
- Data Bits
- Stop Bits
- Parity
- Flow Control
- Read Timeout

Example:

```
Device Type : Weighing Scale
Connection  : RS232
COM Port    : COM3
Baud Rate   : 9600
Data Bits   : 8
Stop Bits   : 1
Parity      : None
```

20. Weighing Scale — Bluetooth

The Local Device Gateway shall support Bluetooth-enabled weighing scales.

It shall support:

- Device discovery
 - Device identification
 - Pairing
 - Connection
 - Data reception
 - Connection monitoring
 - Reconnection
-

21. Weight Data

The system shall support:

- Gross weight
- Tare weight
- Net weight
- Unit
- Stable/unstable status where supported

Formula:

Net Weight = Gross Weight – Tare Weight

Only valid/stable readings should normally be accepted.

22. Weighing Scale Data Parser

Different manufacturers may transmit different formats.

Examples:

```
ST,GS,002850kg  
+002850.0 kg  
WT:2850.0
```

The Device Gateway shall convert vendor-specific messages into a standard format.

The core application shall remain independent of the manufacturer's protocol.

23. Milk Analyser Integration

The system shall support milk analysers through:

- RS232
- Bluetooth

The analyser may provide:

- FAT
- SNF
- CLR
- Density
- Temperature
- Added Water
- Protein
- Lactose
- Other available parameters

24. Milk Analyser — RS232

Configurable parameters:

- COM Port
- Baud Rate
- Data Bits
- Stop Bits
- Parity
- Flow Control
- Timeout
- Data format
- Parsing rules

25. Milk Analyser — Bluetooth

The Local Device Gateway shall support:

- Device discovery
- Pairing
- Connection
- Data reception
- Connection monitoring
- Automatic reconnection

26. Milk Analyser Data Parsing

Example device message:

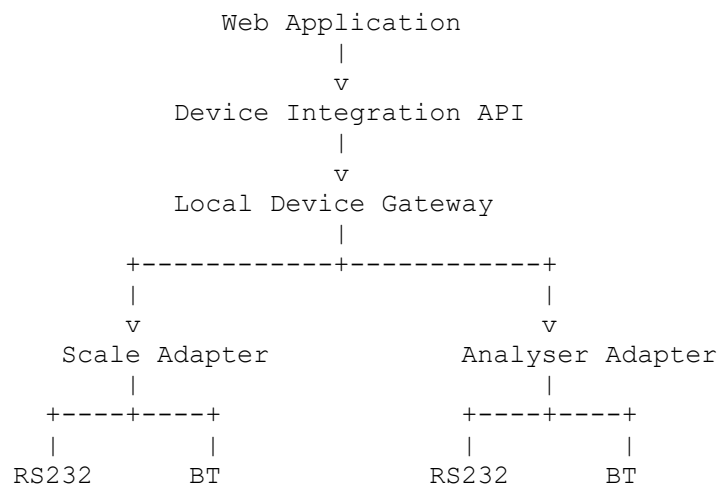
FAT=4.20;SNF=8.50;CLR=28.0;TEMP=7.2

The gateway shall normalize it to:

FAT = 4.20 %
SNF = 8.50 %
CLR = 28.00
Temperature = 7.2 °C

27. Device Abstraction Layer

The main application shall not directly depend on a particular equipment manufacturer.



New equipment should be added through device adapters.

28. Device Configuration

Authorized users shall configure:

General

- Device Name
- Device ID
- Manufacturer
- Model
- Device Type
- Chilling Centre
- Status

RS232

- COM Port
- Baud Rate
- Data Bits
- Stop Bits
- Parity
- Flow Control
- Timeout

Bluetooth

- Device Name
- MAC Address where available
- Pairing Status
- Connection settings

Parser

- Message format
- Field mappings
- Data types
- Unit conversion

Only users with the appropriate device-management permission shall access these functions.

29. Device Test

Authorized technical users shall be able to test devices.

Example:

Weighing Scale

[Test Connection]

✓ Connected

✓ Data Received

Weight: 1250 KG

Stable: YES

Milk analyser:

Milk Analyser

[Test Connection]

✓ Connected
✓ Data Received

FAT: 4.20
SNF: 8.50
CLR: 28.00
Temperature: 7.2°C

30. Device Status

The dashboard shall display device connectivity according to user permissions.

Example:

DEVICE STATUS

Weighing Scale
Connection: RS232
Status: ● Connected

Milk Analyser
Connection: Bluetooth
Status: ● Connected

Internet
Status: ● Connected

31. Automatic Reconnection

The Local Device Gateway shall attempt automatic reconnection after connection loss where technically supported.

The system shall:

1. Detect connection loss.
 2. Record the event.
 3. Notify authorized users.
 4. Attempt reconnection.
 5. Update status.
-

32. Device Communication Logging

The system shall maintain logs containing:

- Device
- Timestamp

- Connection type
- Connection status
- Raw message where appropriate
- Parsed result
- Error
- Reconnection events

Access to technical communication logs shall be RBAC controlled.

33. Source Management

The system shall maintain milk sources such as:

- Milk Collection Centre
- Milk Society
- BMC
- MCC
- Supplier

Information:

- Source ID
 - Source Name
 - Location
 - Contact
 - Milk Type
 - Status
-

34. Vehicle Management

Vehicle information shall include:

- Vehicle number
 - Tanker number
 - Driver
 - Driver mobile
 - Capacity
 - Status
-

35. Milk Reception Transaction

The operator shall create a milk reception transaction.

Required information:

- Transaction number
- Date
- Time
- Source
- Vehicle/tanker
- Milk type
- Operator
- Quantity
- Quality parameters
- Temperature
- Acceptance status

The transaction number shall be automatically generated.

36. Automatic Transaction Association

The weighing-scale and analyser readings shall be associated with the same transaction.

Example:

Transaction: MCC-2026-000123

Source: Collection Centre A
Vehicle: KA01AB1234

Quantity:
2450 KG
Source: Weighing Scale
Connection: RS232

Quality:
FAT: 4.20 %
SNF: 8.50 %
CLR: 28.00
Temperature: 7.2 °C

Source: Milk Analyser
Connection: Bluetooth

Status:
ACCEPTED

37. Quality Validation

Quality limits shall be configurable.

Example:

Parameter	Minimum	Maximum
FAT	3.0	6.0
SNF	8.0	10.0
Temperature	0°C	10°C

The system shall automatically compare analyser readings against configured limits.

38. Milk Acceptance Status

The system shall support:

- ACCEPTED
- REJECTED
- HOLD

The reason for rejection or hold shall be recorded.

39. Manual Override

Only users with the appropriate **Override** permission may override an automatic decision.

The system shall capture:

- Original status
 - New status
 - User
 - Date/time
 - Reason
-

40. Manual Device Fallback

If a device is unavailable, an authorized user may enter readings manually.

The transaction shall record:

Reading Source = MANUAL

rather than **DEVICE**.

The permission to perform manual entry shall be separately configurable.

41. Receipt

After successful completion, the system shall generate an acknowledgement/receipt containing:

- Transaction number
 - Date/time
 - Source
 - Vehicle
 - Quantity
 - FAT
 - SNF
 - CLR
 - Temperature
 - Acceptance status
 - Operator
-

42. Dashboard

The Chilling Centre Manager shall have a dashboard.

Example:

TODAY'S MILK RECEPTION

Total Quantity	25,450 KG
Transactions	18
Accepted	16
Rejected	1
On Hold	1
Average FAT	4.18 %
Average SNF	8.54 %
Average Temperature	6.8 °C

Dashboard data shall respect the user's RBAC and centre-level permissions.

43. Reports

The system shall provide:

Daily Milk Reception

- Transaction
- Date/time
- Source
- Vehicle
- Quantity
- FAT
- SNF
- CLR
- Temperature
- Status

Source-wise

- Source
- Number of loads
- Total quantity
- Average FAT
- Average SNF

Vehicle-wise

- Vehicle
- Number of trips
- Quantity
- Quality

Quality

- FAT
- SNF
- CLR
- Temperature
- Exceptions
- Rejected loads

Device

- Device status
- Connection failures
- Communication errors

Reports shall support:

- Excel
- CSV
- PDF

Report access and export shall be controlled through RBAC.

44. Offline Operation

The system shall support temporary internet outages.

The Local Device Gateway shall buffer transactions locally when the cloud is unavailable.

Example:

Internet: OFFLINE

Transactions Pending Sync: 12
Device Status: Connected

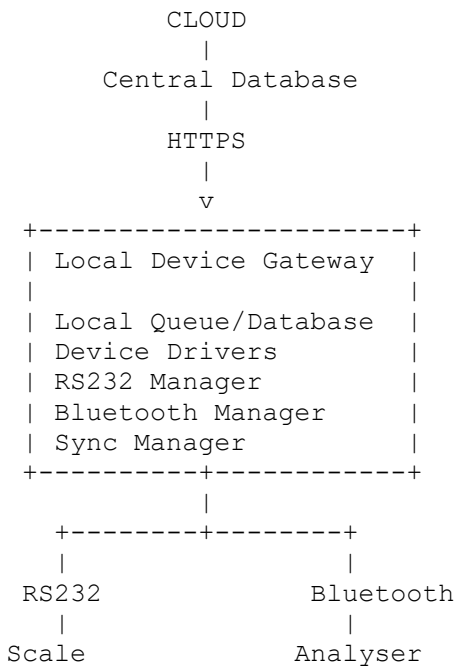
When connectivity returns:

Internet: ONLINE

Synchronization:
12/12 Completed

The synchronization mechanism shall prevent duplicate transactions.

45. Recommended Offline Architecture



46. Security

The system shall implement:

- Secure authentication
- RBAC
- Backend authorization
- HTTPS
- TLS encryption
- Secure API authentication
- Secure credential storage
- Session management
- Audit logging
- Centre-level data isolation

The local gateway shall never expose physical devices directly to the public internet.

47. Secure Communication Architecture

The recommended communication path is:

```
DEVICE
  ↓
LOCAL DEVICE GATEWAY
  ↓
SECURE OUTBOUND HTTPS / WSS
  ↓
CLOUD API
  ↓
CLOUD DATABASE
```

The cloud application shall not require direct access to the local RS232 port.

48. Audit Trail

The system shall maintain an audit trail for:

- Login/logout
- Transaction creation
- Transaction modification
- Transaction cancellation
- Manual quantity entry
- Manual quality entry
- Acceptance override

- Rejection override
- Quality-rule changes
- Device configuration changes
- User creation
- Role changes
- Permission changes
- Source changes
- Vehicle changes

Audit records shall include:

- User
- Role
- Date/time
- Action
- Resource
- Old value
- New value
- Reason where applicable
- Centre

Audit-log access shall itself be RBAC controlled.

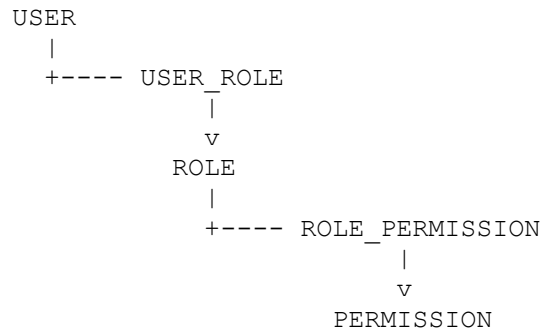
49. Data Model

Minimum entities:

- ChillingCentre
 - Source
 - Vehicle
 - User
 - Role
 - Permission
 - UserRole
 - RolePermission
 - MilkReception
 - MilkQuality
 - Device
 - DeviceConfiguration
 - DeviceReading
 - QualityRule
 - Shift
 - RejectionReason
 - AuditLog
 - SynchronizationLog
-

50. RBAC Data Model

A flexible RBAC model should use:



Example permissions:

MILK_RECEPTION_VIEW
MILK_RECEPTION_CREATE
MILK_RECEPTION_EDIT
MILK_RECEPTION_CANCEL
MILK_RECEPTION_OVERRIDE

QUALITY_VIEW
QUALITY_CONFIGURE

DEVICE_VIEW
DEVICE_CONFIGURE
DEVICE_TEST

REPORT_VIEW
REPORT_EXPORT

USER_VIEW
USER_CREATE
USER_EDIT

ROLE_VIEW
ROLE_CREATE
ROLE_EDIT

AUDIT_VIEW

This allows the organization to create custom roles without changing application code.

51. Centre-Level RBAC

Permissions shall be evaluated together with centre assignment.

Example:

User:
Operator001

Role:
Milk Reception Operator

Centre:
Chilling Centre A

Permissions:
Milk Reception - Create
Milk Reception - View
Receipt - Print

Cannot:
Access Chilling Centre B
Change Quality Rules
Configure Devices
Manage Users
Override Rejection

52. Privileged Operation Controls

The following shall require elevated permissions:

- Quality rule modification
- Device configuration
- Device registration
- Device removal
- Manual quantity entry
- Manual quality entry
- Transaction cancellation
- Transaction modification
- Acceptance override
- Rejection override
- User creation
- Role creation
- Permission assignment
- Report export where sensitive

53. Non-Functional Requirements

Performance

A normal milk reception transaction should be processed within a few seconds after valid device readings are received.

Availability

The cloud application should target high availability.

Local device operation shall continue during temporary internet outages.

Scalability

The system shall support:

- Multiple users
- Multiple chilling centres
- Multiple devices
- Multiple device vendors
- High transaction volumes

Maintainability

New device models should be added through device adapters rather than modifying the core application.

54. Device Integration Specification

Before integrating any physical device, the following information shall be collected:

- Manufacturer
- Model
- Firmware version
- Communication interface
- RS232 protocol
- Baud rate
- Data bits
- Stop bits
- Parity
- Sample raw messages
- Message format
- Commands
- Response format
- Bluetooth protocol/profile
- Pairing mechanism
- SDK/API if available

Each supported device model shall have its own documented integration profile.

55. MVP

Web Application

- Login
- RBAC
- Dashboard
- Milk Reception
- Source Management
- Vehicle Management
- Device Management
- Quality Configuration
- Reports
- Audit Trail

Local Device Gateway

- RS232 support
- Bluetooth support
- Weighing Scale Adapter
- Milk Analyser Adapter
- Data Parsing
- Device Status
- Automatic Reconnection
- Local Buffering
- Cloud Synchronization

Business Functionality

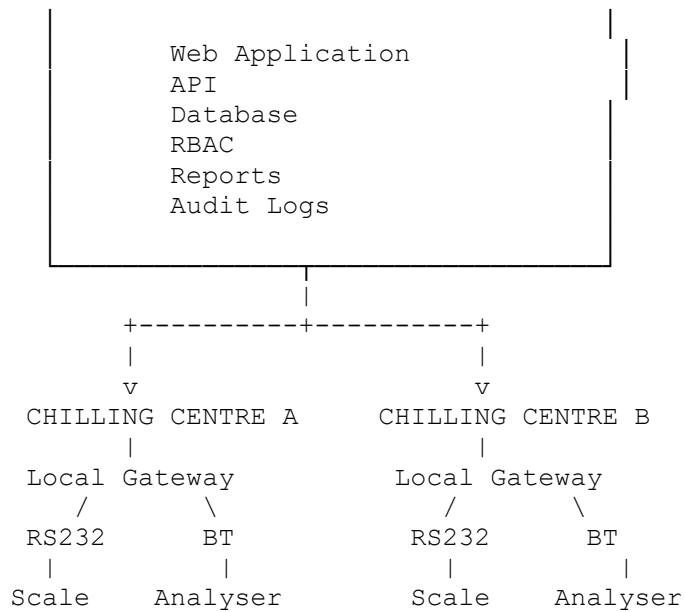
- Quantity Capture
- Quality Capture
- Quality Validation
- Acceptance/Rejection/Hold
- Manual Fallback
- Receipt Generation
- Daily Reports

56. Recommended Deployment Architecture

A single cloud application shall support multiple chilling centres.

CLOUD





57. End-to-End Example

A tanker arrives at the chilling centre.

Step 1 — Authentication

The operator logs into the cloud web application.

The system verifies:

- User
- Password
- Role
- Centre assignment
- Permissions

Step 2 — Start Reception

The operator selects:

Source: Collection Centre A

Vehicle: KA01AB1234

Step 3 — Quantity Measurement

Milk is weighed.

The weighing scale sends:

2,450 KG

through RS232 to the Local Device Gateway.

The gateway parses and normalizes the reading.

Step 4 — Quality Measurement

The milk analyser sends:

FAT: 4.20

SNF: 8.50

CLR: 28.00

Temperature: 7.2°C

through Bluetooth.

Step 5 — Cloud Transaction

The readings are associated with the active milk reception transaction.

Step 6 — Validation

The system evaluates the configured quality rules.

Result:

ACCEPTED

Step 7 — Receipt

The operator generates the acknowledgement.

Step 8 — Dashboard

The transaction becomes visible in the dashboard according to the user's permissions.

Step 9 — Audit

The system records:

- Operator
- Centre
- Device readings
- Transaction
- Decision

- Timestamp
-

58. Acceptance Criteria

The solution shall be considered successful when:

Application

1. The application can be accessed through a web browser.
2. The main application is hosted in the cloud.
3. Users can securely authenticate.
4. RBAC controls access to modules and actions.
5. Centre-level access restrictions are enforced.
6. Backend APIs enforce authorization.

Weighing Scale

7. RS232 weighing scales can be integrated.
8. Bluetooth weighing scales can be integrated.
9. Weight readings are captured automatically.
10. Stable readings can be identified where supported.

Milk Analyser

11. RS232 milk analysers can be integrated.
12. Bluetooth milk analysers can be integrated.
13. Quality readings are captured automatically.
14. FAT/SNF/CLR/temperature can be parsed where provided.

Transaction

15. Quantity and quality readings are associated with the same transaction.
16. Quality validation occurs automatically.
17. Milk can be accepted, rejected or held.
18. Authorized users can override decisions.
19. Manual fallback is available only to authorized users.
20. Receipts can be generated.

Gateway

21. The Local Device Gateway can communicate with devices.
22. Device status can be monitored.
23. Device reconnection is supported.
24. Device communication errors are logged.
25. Local buffering is supported.

Security

26. Sensitive operations are protected by RBAC.
27. User actions are audited.
28. Device configuration is restricted.
29. Quality configuration is restricted.
30. Transaction overrides are audited.

Reporting

31. Daily reception reports are available.
32. Source-wise reports are available.
33. Vehicle-wise reports are available.
34. Quality reports are available.
35. Device reports are available.
36. Report export is permission-controlled.

59. Future Extensions

The architecture shall allow future additions such as:

- Additional device manufacturers
- Additional device protocols
- BMC integration
- Tank temperature sensors
- IoT sensors
- RFID
- Barcode/QR
- GPS
- Farmer/source payment
- ERP integration
- Accounting integration
- Mobile application
- Advanced analytics
- AI/GenAI
- Predictive analytics

These shall not be required for the MVP.

60. Final Solution Vision

The final solution should provide a simple experience to the chilling-centre operator:

LOGIN

↓
SELECT SOURCE / VEHICLE
↓
START MILK RECEPTION
↓
WEIGHING SCALE AUTOMATICALLY PROVIDES QUANTITY
↓
MILK ANALYSER AUTOMATICALLY PROVIDES QUALITY
↓
SYSTEM VALIDATES QUALITY
↓
ACCEPT / HOLD / REJECT
↓
GENERATE RECEIPT
↓
TRANSACTION STORED IN CLOUD
↓
DASHBOARD / REPORTS

Behind this simple user experience, the system will consist of:



Core Architectural Principle

Cloud Web Application

Handles business functionality, RBAC, transactions, dashboards, reports and centralized data.

Local Device Gateway

Handles RS232/Bluetooth communication, device protocols, data parsing, local buffering and synchronization.

Physical Devices

Provide the actual quantity and quality measurements.

This architecture keeps the solution **cloud-hosted, centrally manageable and scalable**, while still allowing reliable integration with the existing physical equipment at each chilling centre.